Fitting

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Lew, in his thesis, uses three compound derivative Lorentzians with the following form,

$$\Delta I(B_0) = \Delta I_{CP} \left(\frac{\Delta B^2}{\Delta B^2 + B_0^2} \right),$$

to model his 4H-SiC EDMR spectra. We will be attempting a similar compound-Lorentzian fit on our 2Gauss, 3Volt, 200MHz EDMR spectra. But first, we will derive the fit functional form from first principle spin mechanics.

This derivation mimics the one presented in Lew Chapter 3, but slightly modified.

1 Single-spin dynamics

1.1 Magnetic Moment and Torque

For an electron of charge $q = -e$ and mass m, the magnetic moment $\vec{\mu}$

$$\vec{\mu} = -g_e \mu_B \frac{\vec{S}}{\hbar}, \qquad \mu_B = \frac{e\hbar}{2m}.$$

The torque on the moment in a magnetic field $\vec{B}$ is

$$\vec{\tau} = \frac{d\vec{S}}{dt} = \vec{\mu} \times \vec{B}.$$

Inserting $\vec{\mu}$ and defining the gyromagnetic ratio $\gamma = g_e \mu_B / \hbar$:

$$\frac{d\vec{S}}{dt} = -\gamma \vec{S} \times \vec{B}.$$

1.2 Larmor Precession

For a static field $\vec{B}_0 = B_0 \hat{z}$, the solution is a uniform precession about $\hat{z}$:

$$\vec{S}(t) = \begin{pmatrix} S_\perp \cos(\omega_0 t + \phi) \\ S_\perp \sin(\omega_0 t + \phi) \\ S_z(0) \end{pmatrix}, \quad \omega_0 = |\gamma| B_0.$$

The Larmor condition is when $\omega = \omega_0$; when the external drive matches the mechanical eigenfrequency, energy can be absorbed resonantly.

2 The Bloch Equations

2.1 Classical Magnetisation

Let the bulk magnetisation $\vec{M} = \sum_i \langle \mu_i \rangle / V$. Averaging the single-spin torque and adding the relaxation terms T_1 and T_2 gives

$$\frac{d\vec{M}}{dt} = \gamma \vec{M} \times \vec{B} - \frac{M_x \hat{x} + M_y \hat{y}}{T_2} - \frac{(M_z - M_0)\hat{z}}{T_1}.$$

T_1 drives $M_z \to M_0$ (longitudinal relaxation). T_2 damps transverse coherence. Our EDMR spectra is taken while sweeping a static field $B\hat{z}$ through resonance while applying a continuous transverse RF field of fixed angular frequency ω:

$$\vec{B}(t) = B\hat{z} + 2B_1 \cos(\omega t)\hat{x}.$$

where the factor of 2 is for simplifying algebra later on with the rotating-wave approximation. B_1 is the rms amplitude.

3 Rotating Frame Transformation

Define a frame that rotates *about* $\hat{z}$ at the drive frequency ω. Let $\vec{M}_{\text{lab}}$ be the lab vector and $\vec{M}$ the rotating-frame vector. Then,

$$\left.\frac{d\vec{M}_{\text{lab}}}{dt}\right|_{\text{lab}} = \left.\frac{d\vec{M}}{dt}\right|_{\text{rot}} + \omega \hat{z} \times \vec{M}.$$

Insert this into the Bloch equations defined above to yield

$$\frac{d\vec{M}}{dt} = \gamma \vec{M} \times \vec{B}_{\text{eff}} - \frac{M_x \hat{x} + M_y \hat{y}}{T_2} - \frac{(M_z - M_0)\hat{z}}{T_1},$$

with an effective field of

$$\vec{B}_{\text{eff}} = (B - \omega/\gamma)\hat{z} + B_1 \hat{x}. \tag{1}$$

The effective field was derived by the *rotating-wave approximation*:

$$2B_1 \cos\omega t\hat{x} = B_1(e^{+i\omega t} + e^{-i\omega t})\hat{x}.$$

In the frame co-rotating at $+\omega$, the $e^{+i\omega t}$ piece is static, while the $e^{-i\omega t}$ term oscillates at 2ω. Provided $\omega \gg 1/T_2$, we drop the latter. What remains is the static $B_1\hat{x}$ present in $\vec{B}_{\text{eff}}$.
Now in steady state (not pulsed) EDMR, we set $d\vec{M}/dt = 0$ and let $\Delta\omega = \omega - \gamma B$. Therefore, with the unknowns M_x, M_y, M_z, we get a 3×3 linear system:

$$\begin{cases} 0 = \gamma\left(M_y B_{\text{eff},z} - M_z B_1\right) - \dfrac{M_x}{T_2}, \\ 0 = \gamma\left(M_z B_1 - M_x B_{\text{eff},z}\right) - \dfrac{M_y}{T_2}, \\ 0 = \gamma\left(M_x B_1 - M_y B_1\right) - \dfrac{M_z - M_0}{T_1}. \end{cases}$$

From Equation 1, we see $B_{\text{eff, z}} = -\Delta\omega/\gamma$. Therefore, the first two equations couple only through B_1 and $\Delta\omega$. Solving yields the solutions

$$M_x = \frac{\gamma B_1 T_2 M_0}{1 + (\Delta\omega T_2)^2 + \gamma^2 B_1^2 T_1 T_2}, \tag{2}$$

$$M_y = \frac{\gamma B_1 T_2 (\Delta\omega T_2) M_0}{1 + (\Delta\omega T_2)^2 + \gamma^2 B_1^2 T_1 T_2}, \tag{3}$$

$$M_z = \frac{M_0}{1 + (\gamma B_1)^2 T_1 T_2 + (\Delta\omega T_2)^2}. \tag{4}$$

4 Lorentzian Line Shape

Now is where things get a bit hand-wavy. The derivation from now on isn't fully firm but is a good heuristic for fitting.
Again from Equation 1, we have

$$\vec{B}_{\text{eff}} = B_z\hat{z} + B_1\hat{x}, \qquad B_z \equiv \frac{\Delta\omega}{\gamma}, \quad \Delta\omega = \omega - \gamma B.$$

Inserting this into the Bloch equations at steady state ($dM_z/dt = 0$) and linearising the equations in B_1, we get

$$\dot{M_x} = -\Delta\omega M_y - \frac{M_x}{T_2} + 0 \cdot B_1, \tag{5}$$

$$\dot{M_y} = +\Delta\omega M_x - \frac{M_y}{T_2} + \gamma M_0 B_1. \tag{6}$$

We now solve Equation 5 in the frequency domain. Defining the complex transverse magnetization

$$M_+(t) \equiv M_x(t) + iM_y(t), \qquad B_+(t) \equiv 2B_1 e^{-i\omega t}.$$

Taking the fourier transform and solving yields

$$M_+(\omega) = \gamma M_0 B_1 \frac{1}{1/T_2 - i\Delta\omega}. \tag{7}$$

Now Equation 7 has the form of

$$M_+(\omega) = \chi_0 \frac{1}{\chi(\omega)} B_1,$$

so the *complex transverse susceptibility* is

$$\chi(\omega) = \frac{\chi_0 T_2}{1 - i\Delta\omega T_2}$$

Separating this into its real and imaginary parts yields

$$\chi'(\omega) = \Re\{\chi(\omega)\} = \frac{\chi_0 T_2}{1 + (\Delta\omega T_2)^2} \tag{8}$$

$$\chi''(\omega) = \Im\{\chi(\omega)\} = \frac{\chi_0 T_2^2 \Delta\omega}{1 + (\Delta\omega T_2)^2} \tag{9}$$

Turns out, the energy a spin system absorbs is proportional to the imaginary $\Im$ part. Therefore, any measurable change in EDMR current – a result of energy absorption – is proportional to $\chi''(\omega)$. Therefore, we have

$$\Delta I(B) \propto \frac{\Delta\omega T_2}{1 + (\Delta\omega T_2)^2}$$

Now, letting the half-width at half-maximum

$$\Gamma \equiv \frac{1}{|\gamma| T_2}$$

Inserting this into χ'', and switching to the field axis $\Delta\omega = \gamma(B - B_{\text{res}})$, we get

$$\Delta I(B) \propto \frac{\Gamma^2}{\Gamma^2 + (B - B_{\text{res}})^2}. \tag{10}$$

which is an exact Lorentzian.

5 Lock-in Amplifier

Here, I believe in our EDMR setup, to reduce noise and distinguish the spin-dependent current changes from the background, we add a small *sine*-wave to the static magnetic field. Specifically, there is a small δB modulation in addition to the static applied B_0. The instantaneous field experienced by the spins is then

$$B(t) = B_0 + \delta B \cos \Omega t \qquad (\delta B \ll B_0).$$

We expand the current

$$I(B(t)) \approx I(B_0) + \frac{\partial I}{\partial B}\Big|_{B_0} \delta B \cos \Omega t + \mathcal{O}(\delta B^2)$$

Since EDMR uses a lock-in amplifier, we define the lock-in amplifier's *demodulted* output, the actual observed spectra as

$$S(B_0) = \frac{2}{T_{\text{avg}}} \int_{t_0}^{t_0+T_{\text{avg}}} I(B_0 + \delta B \cos \Omega t) \cos \Omega t \, dt \tag{11}$$

Therefore, if we expand to first-order, we get

$$I(B_0 + \delta B \cos \Omega t) \cdot \cos \Omega t \approx I(B_0) \cos \Omega t + \frac{\partial I}{\partial B}\Big|_{B_0} \delta B \cos^2 \Omega t + \mathcal{O}(\delta B^2)$$

We can solve the integral by just averaging,

$$\langle I(B_0) \cos \Omega t \rangle = I(B_0) \langle \cos \Omega t \rangle = 0$$
$$\langle \cos^2 \Omega t \rangle = \frac{1}{2}$$

Hence,

$$S(B_0) = \frac{2}{T_{\text{avg}}} \int \left[I(B_0) + I'(B_0) \delta B \cos \Omega t \right] \cos \Omega t dt = \frac{\delta B}{2} \frac{\partial I}{\partial B}\Big|_{B_0} \tag{12}$$

The lock-in outputs a current proportional to the **first derivative** of the underlying $I(B)$. This is what is being observed in the EDMR spectra.

6 Markov Chain Monte Carlo (MCMC)

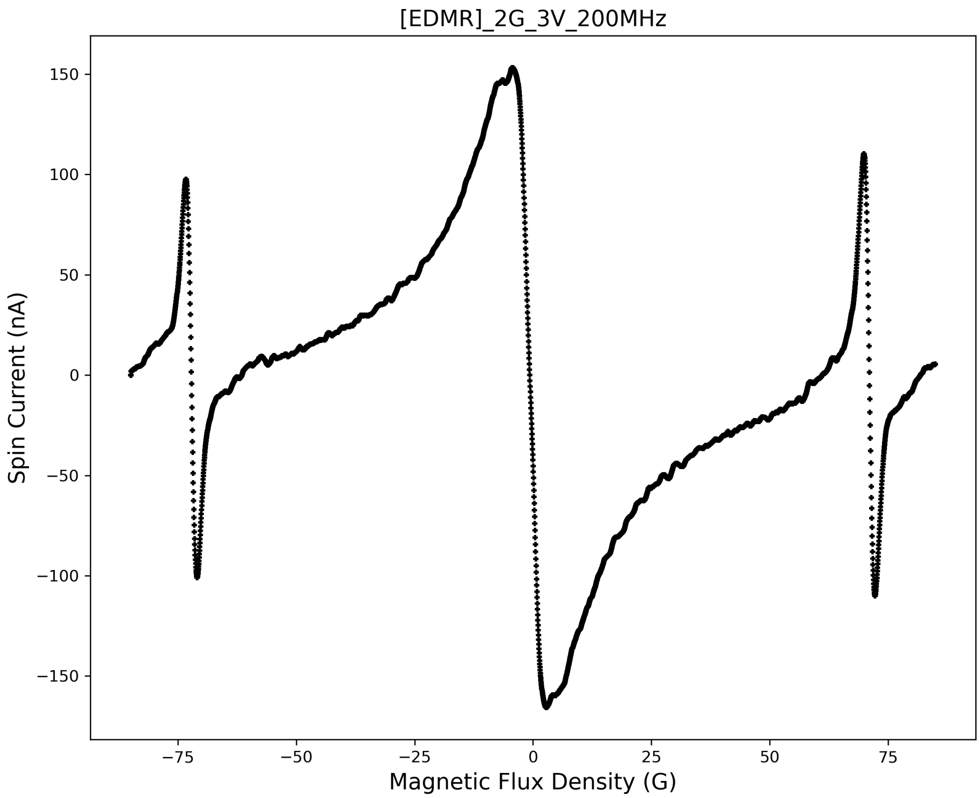

Figure 1: EDMR spectra at 2G, 3V, and 200MHz. It can be modeled via three compound-lorentzian derivatives.

From Equation 10, we have

$$\Delta I(B) = C\frac{\Gamma^2}{\Gamma^2 + (B - B_{\text{res}})^2}, \qquad \Gamma = \frac{1}{|\gamma|T_2}$$

We differentiate the above equation and insert it into Equation 12:

$$S(B) = -2A\frac{B}{(B^2 + \Gamma^2)^2}, \qquad A \equiv \frac{\delta B}{2}C\Gamma^2.$$

This is the first derivative of a centered Lorentzian. Empirically, the EDMR spectra constitutes three independent processes (hyperfine, zero-field, possibly relaxation ?) Therefore the compound model follows

$$S_{\text{model}}(B;\ \theta) = S_0 - 2B \sum_{k=1}^{3} \frac{A_k}{(B^2 + \Gamma_k^2)^2},$$

with parameter vector

$$\theta = (S_0, A_1, A_2, A_3, \Gamma_1, \Gamma_2, \Gamma_3, \sigma).$$

where S_0 is the baseline offset and σ is the stddev of the Gaussian noise. There exists an implicit T_2 within each fitted Γ_k. T_1 does not appear.

6.1 Likelihood

For measured pairs $\{B_i, S_i\}_{i=1}^{N}$, where N is the number of random walkers and independent Gaussian noise:

$$\mathcal{L}(\theta) = \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(S_i - S_{\text{model}}(B_i;\ \theta))^2}{2\sigma^2}\right]$$

Therefore,

$$\ln \mathcal{L} = -N \ln \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^{N} \left[S_i - S_{\text{model}}(B_i;\ \theta)\right]^2 - \frac{N}{2} \ln(2\pi).$$

6.2 Priors

We adopt weakly informative priors without any biases:

$$\begin{aligned} S_0 &\sim \mathcal{U}\left[\min S_i, , \max S_i\right] \\ A_k &\sim \mathcal{N}(0, 200\text{nA}^2) \\ \Gamma_k &\sim \log \mathcal{U}\left[0.05, 100\right] \text{G} \\ \sigma &\sim \text{Half-Cauchy} \end{aligned}$$

The final posterior is then

$$\ln P(\theta|\ \text{data}) = \ln \mathcal{L}(\theta) + \sum_{\xi \in \theta} \ln p(\xi).$$

7 Future work

This derivation is focused on the ±50G region of the spectra and does not include the resonance peaks beyond ±70G. To get a full description of an infinitely-long spectra in gauss-space, we could use a "gauss-series" similar to a time-series and do a compound fourier lorentzian-derivative fit that repeats at every resonance condition dependending on the material. This would be very interesting and doable. But for now, I'll work on coding this main zero-field fit and getting some results.

https://journals.aps.org/prb/abstract/10.1103/PhysRevB.76.054409
https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.130.266703
All-electrical Quantum Sensing with Silicon Carbide Devices – Christopher Tao-Kuan Lew
https://en.wikipedia.org/wiki/Bloch_equations (wikipedia I know)